1. SYSTEM OVERVIEW & ARCHITECTURAL FOUNDATION

N100 Financial Intelligence Platform • Technical Analyst Manual

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The N100 Financial Intelligence Platform is an enterprise-grade equity analysis suite developed for Bluestock Fintech. It automates data extraction, financial statement normalization, KPI computation, multi-factor screening, peer benchmarking, and machine learning clustering for 92 top Indian listed companies in the Nifty 100 universe.

Section Reference: *Chapter 1.0 | Approved by Bluestock Engineering & Analytics Council*

2. DATA PIPELINE & ETL ENGINEERING SPECIFICATIONS

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The ingestion pipeline processes 10 raw Excel and CSV sources into data/nifty100.db via SQLite relational tables. Key engineering features include automated year normalization (converting YYYY-MM and FY dates), company ticker mapping, handling negative base years for CAGR (flagging TURNAROUND), and handling zero interest expenses (flagging DEBT FREE).

Section Reference: *Chapter 2.0 | Approved by Bluestock Engineering & Analytics Council*

3. INSTALLATION, SETUP & DEPENDENCY MANAGEMENT

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To deploy the environment locally:

1. pip install -r requirements.txt
2. make load (ingests all Excel files into SQLite)
3. make ratios (computes all financial ratios across 10 years)
4. make test (executes the 70 unit tests)
5. make dashboard (runs Streamlit on port 8501)
6. make api (runs FastAPI on port 8000).

Section Reference: *Chapter 3.0 | Approved by Bluestock Engineering & Analytics Council*

4. NAVIGATING THE STREAMLIT DASHBOARD — SCREEN 1: MARKET OVERVIEW

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Screen 1 provides high-level executive KPI cockpits: Market Cap distribution, Nifty 100 Index performance, sector weighting treemaps, and live market breadth indicators. Interactive Plotly charts allow filtering by Market Cap category and Broad Sector.

Section Reference: *Chapter 4.0 | Approved by Bluestock Engineering & Analytics Council*

5. NAVIGATING THE STREAMLIT DASHBOARD — SCREEN 2: EQUITY SCREENER

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The Screener screen features configurable sliders for Minimum ROE, Maximum Debt-to-Equity, Minimum Free Cash Flow, and 5-Year CAGR. Pre-built presets ('Quality Compounders', 'Defensive Dividend', 'Value Cyclicals') enable one-click filtering with instant CSV export.

Section Reference: *Chapter 5.0 | Approved by Bluestock Engineering & Analytics Council*

6. NAVIGATING THE STREAMLIT DASHBOARD — SCREEN 3: PEER COMPARISON & RADAR

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Screen 3 compares target companies against sector peers across 8 normalized dimensions: ROE, ROCE, OPM, Revenue CAGR, PAT CAGR, D/E, Interest Coverage, and FCF Yield. Radar charts display the company profile relative to the peer median and benchmark leader.

Section Reference: *Chapter 6.0 | Approved by Bluestock Engineering & Analytics Council*

7. NAVIGATING THE STREAMLIT DASHBOARD — SCREEN 4 & 5: VALUATION & CASH FLOW INTELLIGENCE

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The Valuation module evaluates historical P/E, P/B, and EV/EBITDA multiples relative to 5-year averages. The Cash Flow Intelligence screen audits CFO/PAT conversion quality, CapEx intensity, distress signals (CFO < 0 and CFF > 0), and deleveraging trends.

Section Reference: *Chapter 7.0 | Approved by Bluestock Engineering & Analytics Council*

8. NAVIGATING THE STREAMLIT DASHBOARD — SCREEN 6 & 7: KMEANS CLUSTERING & DOCUMENTS

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Screen 6 visualizes 5 distinct archetypal clusters derived from unsupervised KMeans machine learning. Screen 7 provides an institutional document repository linking annual reports, investor presentations, and audited transcripts.

Section Reference: *Chapter 8.0 | Approved by Bluestock Engineering & Analytics Council*

9. AUTOMATED REPORT GENERATION — TEARSHEETS & SECTOR REVIEWS

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The platform automatically compiles 2-page institutional company tearsheets using ReportLab. Page 1 displays KPI tiles, revenue/profit charts, and ROE/ROCE line graphs. Page 2 details Balance Sheet stacked bars, Cash Flow waterfalls, pros/cons, and capital allocation badges.

Section Reference: *Chapter 9.0 | Approved by Bluestock Engineering & Analytics Council*

10. FASTAPI REST SERVER & API ENDPOINT REFERENCE

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The REST API serves 16 endpoints prefixed with /api/v1:

- GET /api/v1/health: Server health and SQLite table counts
- GET /api/v1/companies: List all 92 companies with sector and returns
- GET /api/v1/screener: Multi-parameter query screener
- GET /api/v1/sectors: 11 sector medians
- Example: curl http://localhost:8000/api/v1/companies/TCS/ratios

Section Reference: *Chapter 10.0 | Approved by Bluestock Engineering & Analytics Council*

11. MACHINE LEARNING ARCHETYPE CLUSTERING METHODOLOGY

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KMeans clustering applies StandardScaler to 5 core financial metrics: ROE %, D/E, Revenue CAGR 5Y, FCF CAGR 5Y, and OPM %. Elbow plot analysis confirms k=5 as optimal. The resulting archetypes are: High-Quality Compounders, Defensive Dividend Payers, Value Cyclicals, Distressed/Turnaround, and Emerging Growth.

Section Reference: *Chapter 11.0 | Approved by Bluestock Engineering & Analytics Council*

12. TROUBLESHOOTING, SYSTEM MAINTENANCE & OPERATIONAL FAQ

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Common operational procedures:

- Database Lock: Delete temporary .db-journal files and restart uvicorn.
- Missing Years in Analysis: Parser automatically logs non-matching entries to output/parse_failures.csv.
- Running Full QA: Run 'pytest tests/ -v' to verify all 70 tests pass with 0 failures.

Section Reference: *Chapter 12.0 | Approved by Bluestock Engineering & Analytics Council*